Operations Research 1: Deterministic Models
Gurobi Setup

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Fall, 2025

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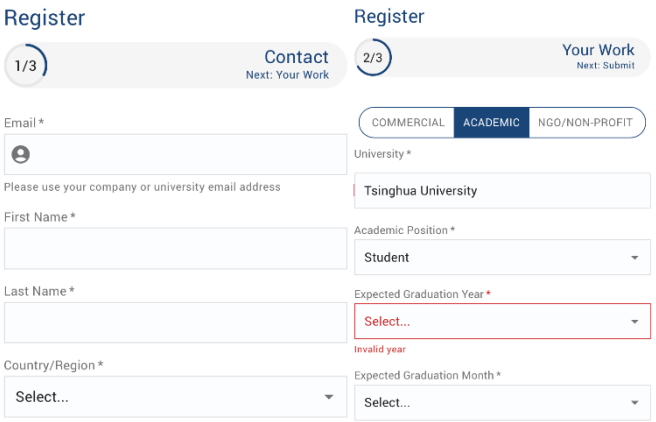
Gurobi and License Installation

Gurobi Installation

- Register with your Tsinghua email account as an academic user.
  - https://pages.gurobi.com/registration



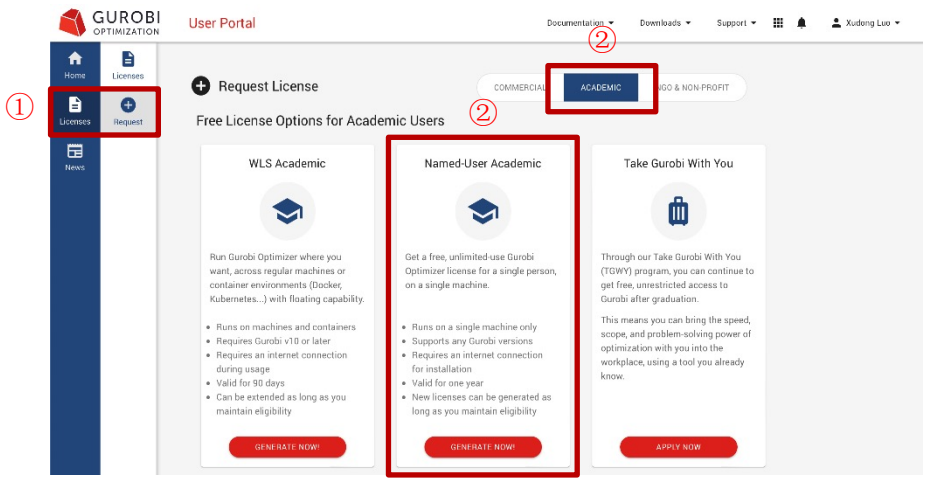
- Login to your account, download the latest version of Gurobi, and install it on your computer.
  - https://www.gurobi.com/downloads/gurobi-software/

Gurobi Optimizer — v12.0.3 Release Notes — md5sums

v12.0.3	Installer	md5 Checksum
x64 Windows	Gurobi-12.0.3-win64.msi	Gurobi-12.0.3-win64.msi.md5
x64 Linux	gurobi12.0.3_linux64.tar.gz	gurobi12.0.3_linux64.tar.gz.md5
macOS Universal2	gurobi12.0.3_macos_universal2.pkg	gurobi12.0.3_macos_universal2.pkg.md5
arm64 Linux	gurobi12.0.3_armlinux64.tar.gz	gurobi12.0.3_armlinux64.tar.gz.md5

Gurobi License Application

- Apply for an academic license.
  - https://portal.gurobi.com/iam/home
  - ① Select “Licenses” and then “Request.”
  - ② Apply for an academic license “Named-User Academic.”



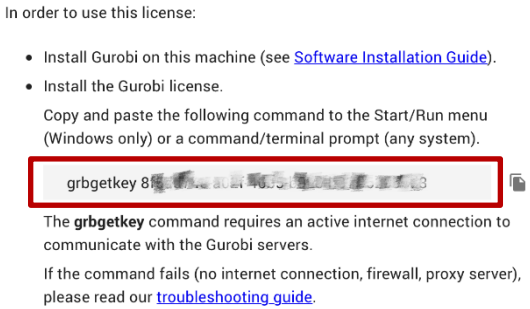
- More details on the application of Gurobi licenses:
  - https://support.gurobi.com/hc/en-us/articles/12684663118993-How-do-I-obtain-a-Gurobi-license#section:ObtainLicense

Network Environment

- When you activate your Gurobi license, you must use the education network corresponding to your registered account.
  - That is, if you register as a student of Tsinghua University, connect to the Tsinghua University network on Tsinghua campus, when you activate your Gurobi license.
  - Note: Gurobi will check whether you are an academic user based on the IP address of your education network.

Gurobi License Installation

- Copy the license installation command of your academic license from the Gurobi webpage.


In order to use this license:

- Install Gurobi on this machine (see Software Installation Guide).
- Install the Gurobi license.

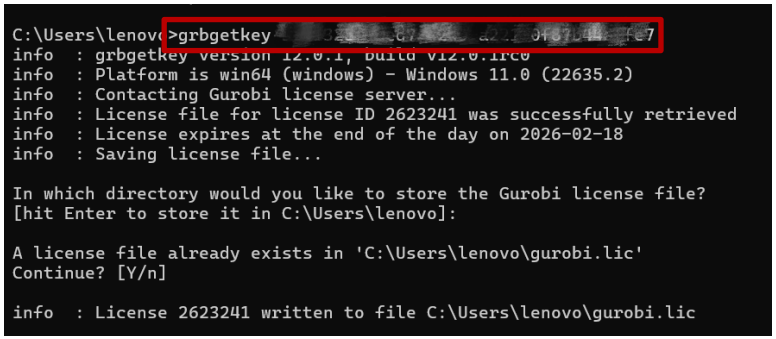
  Copy and paste the following command to the Start/Run menu (Windows only) or a command/terminal prompt (any system).

  grbgetkey 8[redacted]3

  The **grbgetkey** command requires an active internet connection to communicate with the Gurobi servers.

  If the command fails (no internet connection, firewall, proxy server), please read our troubleshooting guide.

- Paste the license installation command in Anaconda Prompt (Windows OS) or Terminal (Mac OS or Linux OS) and press Enter. The default setting should be fine.


```
C:\Users\lenovo>grbgetkey [redacted]fe7
info  : grbgetkey version 12.0.1, build v12.0.1rc0
info  : Platform is win64 (windows) - Windows 11.0 (22635.2)
info  : Contacting Gurobi license server...
info  : License file for license ID 2623241 was successfully retrieved
info  : License expires at the end of the day on 2026-02-18
info  : Saving license file...

In which directory would you like to store the Gurobi license file?
[hit Enter to store it in C:\Users\lenovo]:

A license file already exists in 'C:\Users\lenovo\gurobi.lic'
Continue? [Y/n]

info  : License 2623241 written to file C:\Users\lenovo\gurobi.lic
```

Test the Installation of Gurobi License

- Enter the following command in Anaconda Prompt (Windows OS) or Terminal (Mac OS or Linux OS).

  `gurobi_cl`

- If the output matches the snapshot below, you have successfully installed Gurobi and the academic license on your computer.
  - Note: There may be slight differences between versions.

```
C:\Users\lenovo>gurobi_cl
Set parameter Username
Set parameter LicenseID to value 2623241
Set parameter LogFile to value "gurobi.log"
Using license file C:\Users\lenovo\gurobi.lic
Academic license - for non-commercial use only - expires 2026-02-18

Usage: gurobi_cl [--command]* [param=value]* filename
Type 'gurobi_cl --help' for more information.
```

gurobipy Installation

gurobipy Installation

- Now you have installed Gurobi successfully, but you still need a “bridge” to connect Gurobi and Python, that is, gurobipy.
- Installation options
  - Download and install it from PyPI through pip.
  - Install it as part of the Gurobi conda package.
  - More details on the installation options of gurobipy
    - https://support.gurobi.com/hc/en-us/articles/360044290292-How-do-I-install-Gurobi-for-Python
- We will use the first option as an example.

gurobipy Installation

- Download and install it from PyPI through pip.
  - Open Anaconda Prompt (Windows OS) or Terminal (Mac OS or Linux OS) as administrator.

```
C:\Users\lenovo>
```

  - Enter the following command.

```
python -m pip install gurobipy
```

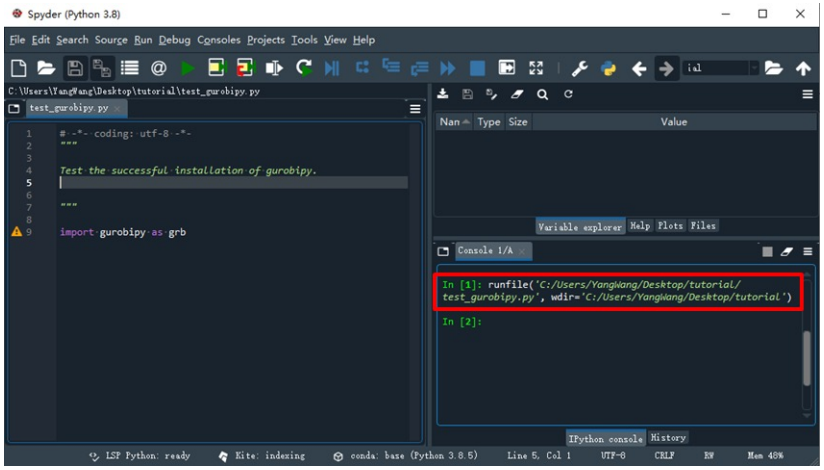
```
C:\Users\lenovo>python -m pip install gurobipy
Collecting gurobipy
  Using cached gurobipy-12.0.1-cp39-cp39-win_amd64.whl (11.2 MB)
Installing collected packages: gurobipy
Successfully installed gurobipy-12.0.1
```

Test the Installation of gurobipy

- Enter the following code in Spyder.

```
import gurobipy as grb
```

- Save the program to your working directory.
  - E.g., name the program file, such as “test_gurobi.py.”
- Run the program.
  - If the output (in the red box) does not show any error, you have successfully installed gurobipy on your computer.



Test Run

Test Run: Wyndor Glass Co.

- Run main.py
  - If the output (in the red box) shows the optimal solution and objective function value, you have successfully used Gurobi to formulate and solve the problem.

```
Optimize a model with 3 rows, 2 columns and 4 nonzeros
Model fingerprint: 0xee003ec0
Coefficient statistics:
  Matrix range     [1e+00, 3e+00]
  Objective range  [3e+03, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [4e+00, 2e+01]
Presolve removed 2 rows and 0 columns
Presolve time: 0.00s
Presolved: 1 rows, 2 columns, 2 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0    4.5000000e+04   1.500000e+00   0.000000e+00      0s
       1    3.6000000e+04   0.000000e+00   0.000000e+00      0s

Solved in 1 iterations and 0.00 seconds (0.00 work units)
Optimal objective  3.600000000e+04

The optimal solutions:
    Doors: 2.0
    Windows: 6.0
The optimal objective function value: 36000.0
```

Further Information

- In the next tutorial, we will talk about the mechanisms of these two files and how they exactly work.
- You are encouraged to read the code and see if you can understand some parts of it.
- Feel free to browse the Gurobi website and get familiar with Gurobi.
  - https://www.gurobi.com/resource/starting-with-gurobi/